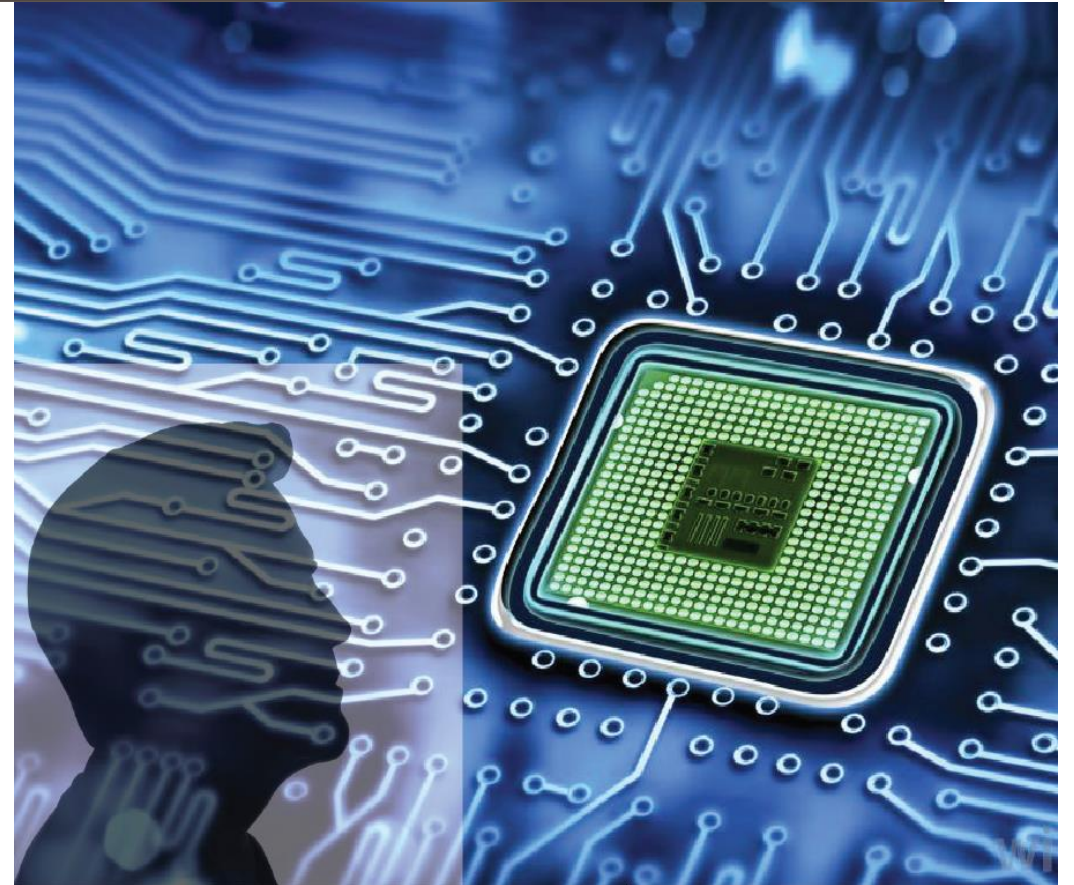


VLSI DESIGN FLOW

Week-2, Lecture-1 Introduction

Sneh Saurabh
19th August, 2024

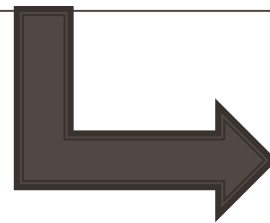


VDF: Revision/Announcements

- Project Topics

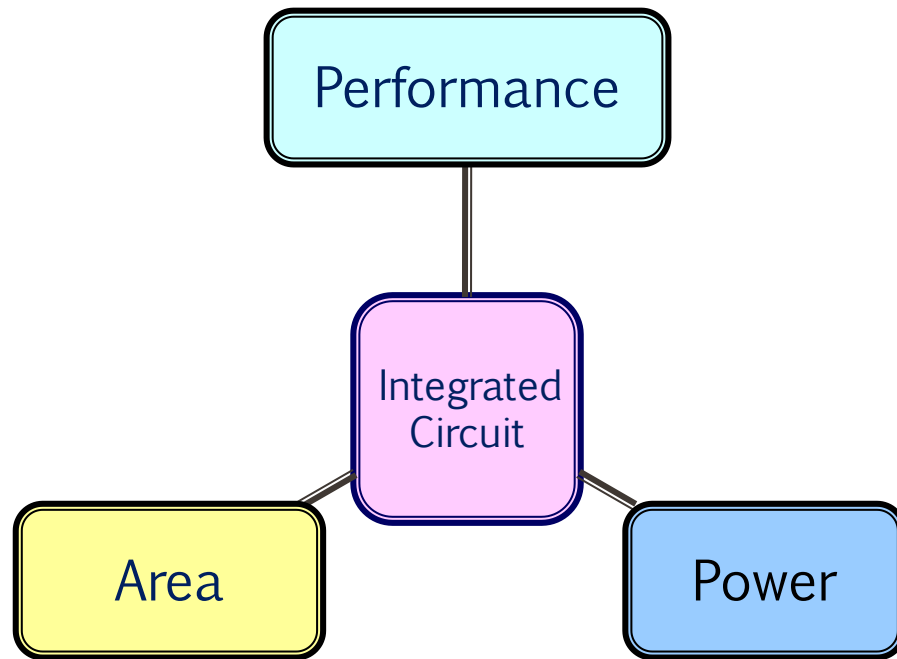


Introduction: Some Basic Concepts



Figures of Merit

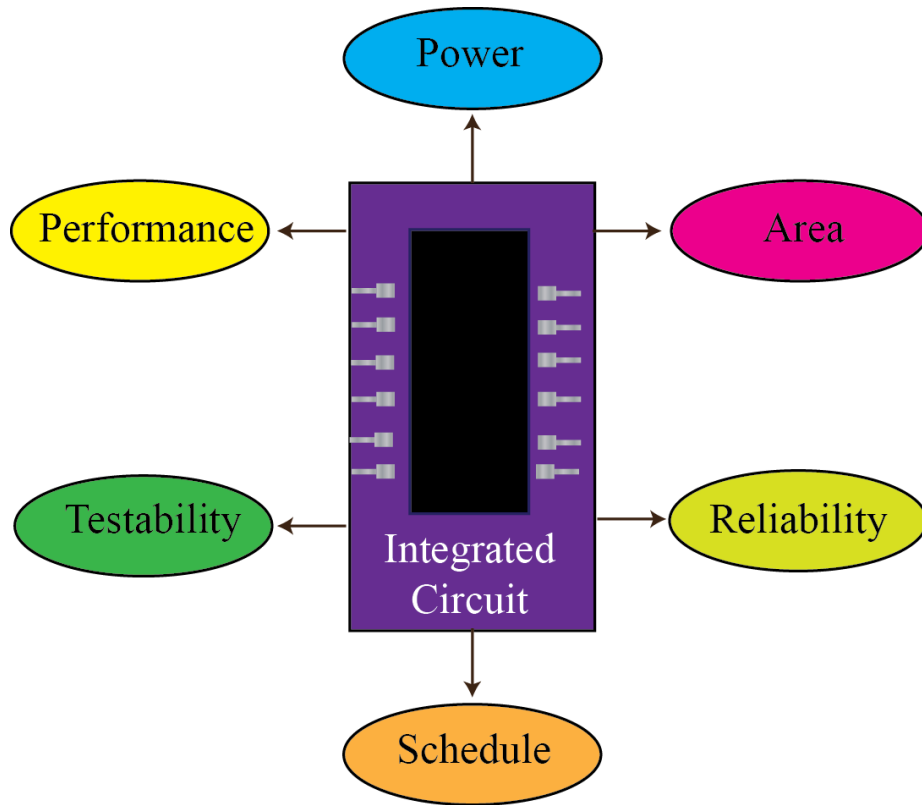
Figures of Merit (FoMs) (1)



Quality Metrics

- How do you assess the “*goodness*” of an IC?
- **Power, Performance, Area (PPA) measure**
 - Power: sum of static and dynamic power consumed by the IC
 - Performance: Maximum frequency of clock at which the IC will work
 - Area: area of the die of the IC
- Example: (1 W, 1.1 GHz, 1 mm²)

Figures of Merit (FoMs) (2)



- PPA also called Quality of Result (QoR)
- Improving one measure might adversely effect other measure(s)
 - Some measures might be required to be traded-off

- Other FOMs:
 - Testability
 - Reliability
 - Schedule

- Mathematical optimum FoM for a given design is rarely known or achieved
- Goal of a design flow is to find one of the feasible solutions with acceptable FoM

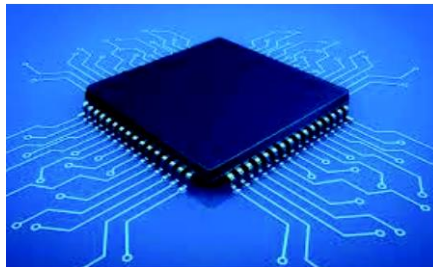
VLSI Design Flow: A top-level perspective

Chip Designing: Input and Output



Idea

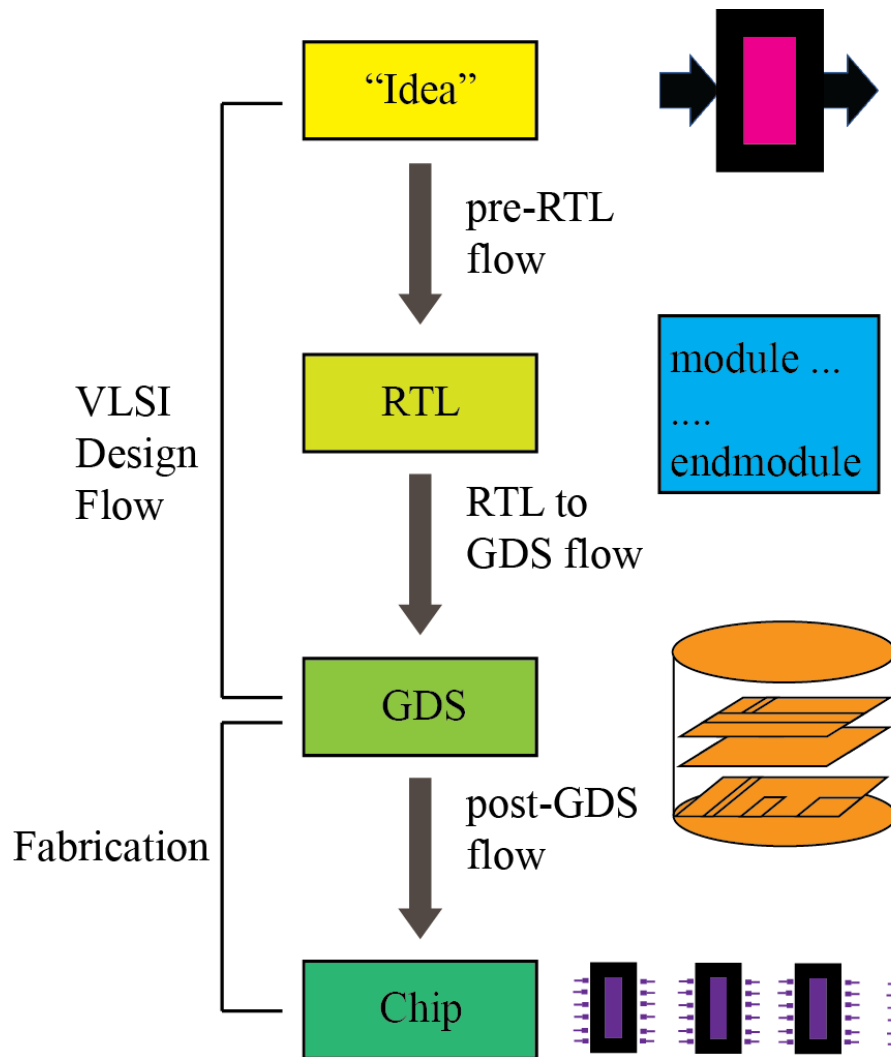
How to tackle this problem?



Chip

Divide and conquer

VLSI Design Flow: Divide and Conquer



RTL: Register Transfer Level (Verilog, VHDL)

GDS: Graphical Database System (Layout)

Idea to RTL Flow: takes a high-level idea/concept of a product and represents the hardware portion of the implementation in RTL.

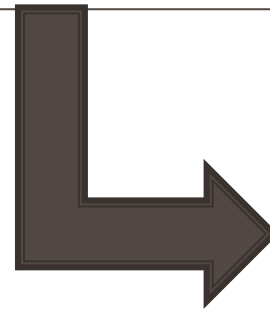
RTL to GDS flow: takes an RTL through various stages of logical and physical design steps and finally represents the design as GDS.

GDS to Chip Flow: takes a GDS, prepares masks for a given GDS and fabricates/tests/packages chips

VLSI Design Flow: Design vs. Fabrication

Flow	Designing/Fabricating
Idea to RTL Flow	<i>Designing</i> at the level of System, Architecture and Behavior of the circuit.
RTL to GDS flow	<i>Designing</i> at the level of Logic, Gates, Transistors, Layout and Devices
GDS to Chip flow	<i>Fabrication</i> and <i>Manufacturing</i> of the chip

VLSI Design Flow: A top-level perspective



Abstraction

Abstraction: Basic Concept

Abstraction:

- Hiding lower level details in a description
- VLSI design flow: At each subsequent design step, details are added to the design and abstraction decreases

	Level of Abstraction	Representation
Idea to RTL Flow	Very high	System, Behavior
RTL to GDS flow	Decreases subsequently down the flow	RTL, Gate, Transistor, Layout
GDS to Chip flow	No abstraction, Actual Implementation	Mask, Integrated Circuit.

Why to Abstract?

Optimization: Choosing right combination of design parameters to obtain desired QoR by trading-off some of them.

Turn-around Time: Time taken to make changes in a design

- ✓ At higher level of abstraction large number of solutions can be analyzed in less amount of time.
- ✓ Result of optimization at the higher level of abstraction is expected to be better.

	Scope of Optimization	Turn-around time
Idea to RTL Flow	Very High	Low
RTL to GDS flow	Decreases subsequently down the flow	Increases subsequently down the flow
GDS to Chip flow	No optimization. Some correction.	Very costly re-spin

Abstraction: Illustration

Consider that the functionality is represented in two ways:

- A. Logic Formula: $F = (A + B)'$
- B. Using a standard cell delivering NOR function placed and connected on the layout.

Which of the above representations:

- 1. Has greater abstraction?

Ans: A

- 2. Smaller turn-around time in evaluating different implementations?

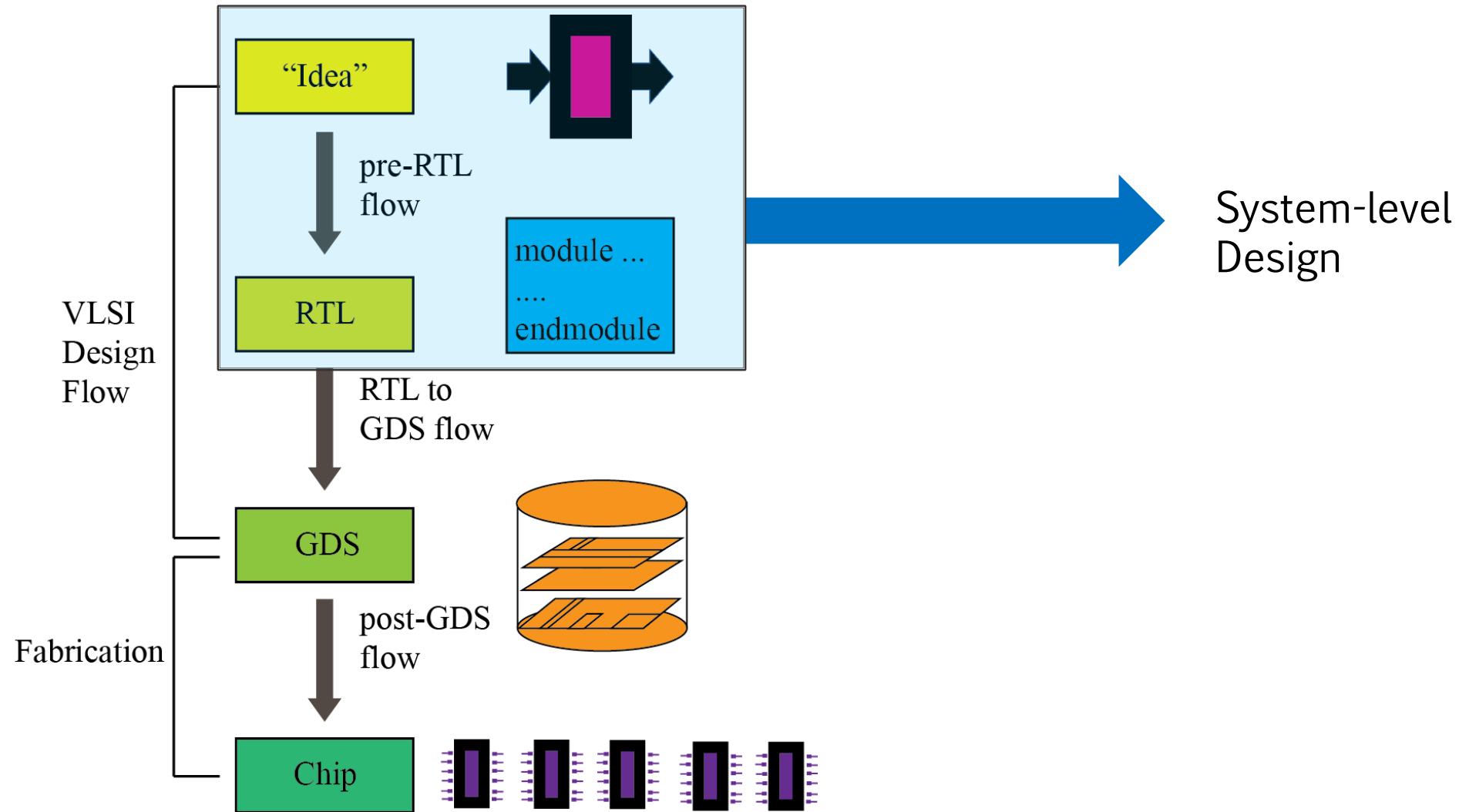
Ans: A

- 3. Greater accuracy in evaluation?

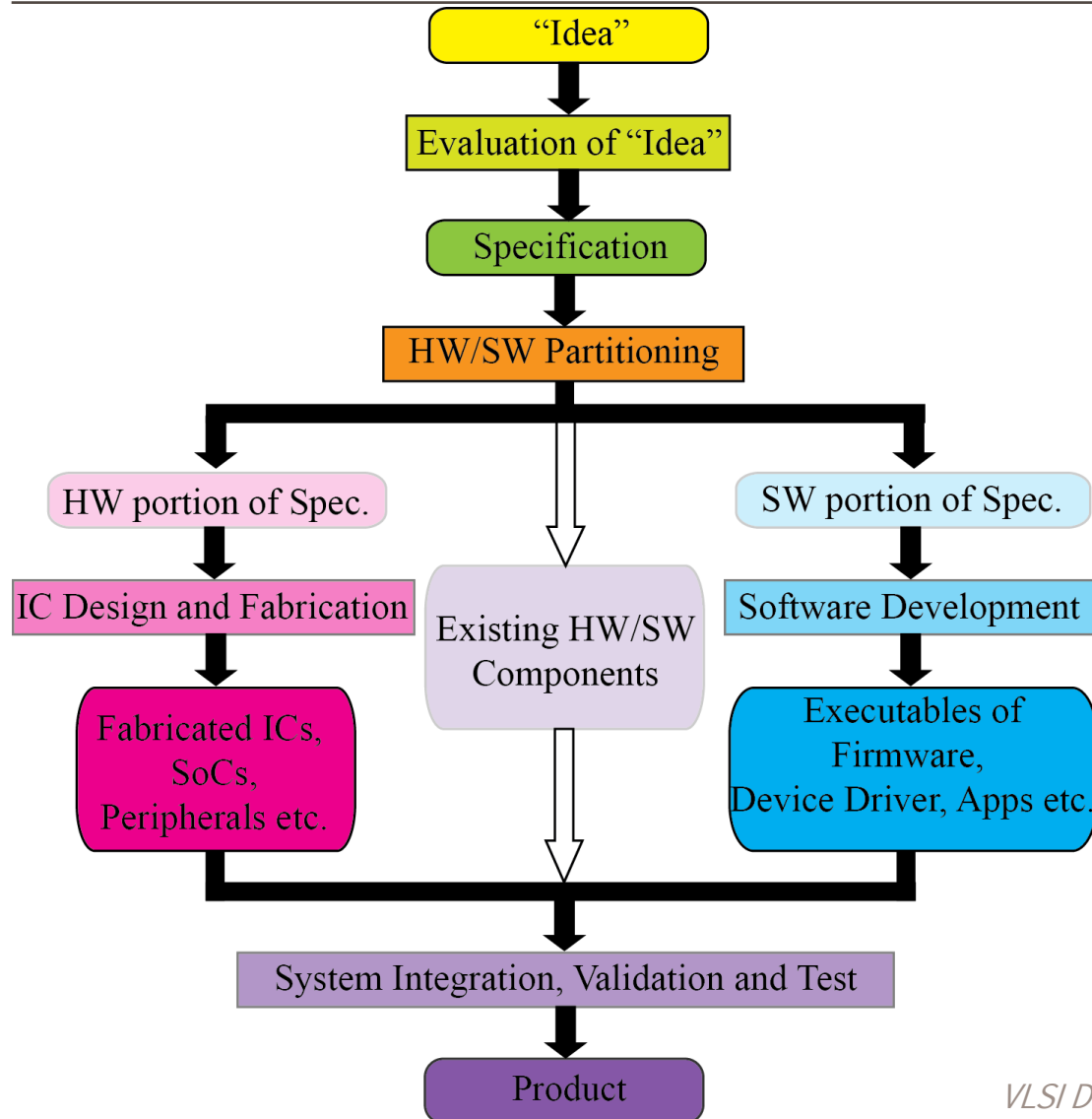
Ans: B

Pre-RTL Methodologies

Pre-RTL Methodologies



System-level Design: Top View



Evaluation of "idea":

- Market requirement
- Financial viability
- Technical feasibility

Preparing specifications:

- Features (functionality)
- PPA
- Time to market (TTM)

HW—SW Partitioning:

- Identify components
- Determine which components to implement in HW/SW

- HW/SW Development (separately)
- System Integration, Validation, Test
- Final Product

References

- S. Saurabh, “Introduction to VLSI Design Flow”. Cambridge: *Cambridge University Press*, 2023.